PROJECT AEGIS: UPI SYSTEM & MASTER FLOWCHART ARCHITECTURE

Complete Technical Blueprint: Dedicated UPI 2.0 Pre-PIN Defense & Whole-Project Unified Flowchart

Document ID: AEG-FLOWCHART-2026	**Classification:** Confidential / Payment Flowchart	**Date:** September 2026
Target Rails: NPCI UPI 2.0, CBS, ISO 20022	**Engines:** Pre-PIN Shield, Ghost Sandbox, Velocity	**Status:** Production Architecture

1. DEDICATED UPI 2.0 PRE-PIN SHIELD & PAYMENT RAIL ARCHITECTURE FLOWCHART

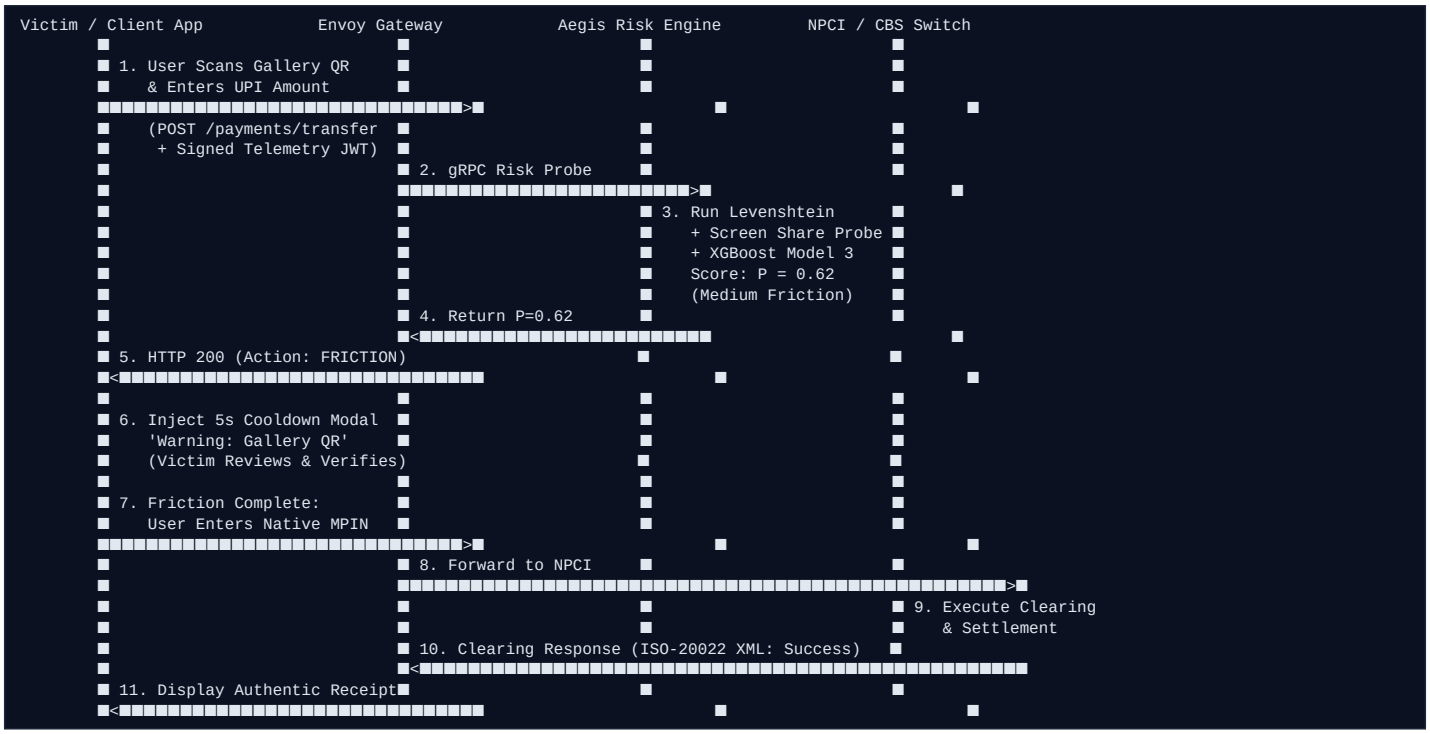
This flowchart specifies the end-to-end processing pipeline for UPI 2.0 transactions, illustrating how optical Camera2 vs gallery image isolation, Levenshtein typo-squatting scanning, OS-level call/mirror hooks, and the Envoy Upstream Gateway work together to intercept fraudulent payments **prior to MPIN entry**.

2. WHOLE PROJECT MASTER UNIFIED SYSTEM ARCHITECTURE FLOWCHART

This master architecture diagram illustrates how the three operational vectors—**Authentication & Honeypot**, **UPI 2.0 Payment Rail**, and **Card Velocity Defense**—converge into a unified distributed threat intelligence and enforcement architecture.

3. TECHNICAL SEQUENCE DIAGRAM: UPI PRE-PIN INTERCEPTION VS SETTLEMENT

Below is the message-level sequence diagram detailing how telemetry tokens, gateway evaluation, dynamic friction, and settlement operate across time:



4. Complete Subsystem Capability & Execution Matrix:

Subsystem	Trigger Event	Latency SLA	Mathematical Engine	Enforcement Output
UPI Pre-PIN Shield	Target VPA input or QR scan	< 15 ms	Levenshtein edit distance & Model 3	Block P>0.70 or Friction 5s
Ghost Sandbox Pod	3 failed passes + H < 0.50	Ephemeral (30m)	Shannon Entropy H = -∑p*log2(p)	Synthetic ■84,250 & STUN harvest
Card Velocity Guard	Dual charges at distant coords	< 2.0 ms	Haversine d = 2R*asin(√a)	Auto-freeze if v > 900 km/h
Mule Graph Engine	Intercepted beneficiary VPA	< 15 ms	Neo4j Cypher 1-3 hop fan-out	Proactive freeze & 1930 dispatch